

Use-Case Flow Specification

5. Academic Elective Bidding & Allocation System

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Use Case: UC-03 — Submit Bid Preferences and Run Allocation

Primary Actor: Student | **Supporting Actors: Academic Registrar, Timetable System**

Preconditions

- Student is authenticated via university SSO and has an active elective bidding window open.
- Student has an available balance of 100 bidding credits and has not already submitted a bid for this cycle.
- The elective catalogue, seat capacities, and timetable slots have been published by the Academic Registrar.

Postconditions

- Success: The student's ranked preferences and credit distribution are recorded, prerequisites are validated, and the bid is queued for the allocation solver.
- Failure: No bid record is created, and the student's credit balance remains unchanged.

Main Success Scenario

1. Student selects "Elective Bidding" from the academic portal. System displays the list of available electives with seats remaining and prerequisite tags.
2. Student ranks between 3 and 8 electives in order of preference and distributes exactly 100 bidding credits across them.
3. System validates that the total allocated credits equal 100 and that no elective appears twice in the ranking.
4. System checks the student's academic record and confirms all prerequisite courses for the selected electives are completed (UC-02: Validate Prerequisites, included).
5. Student confirms and submits the bid. System timestamps and stores the bid, then deducts the 100 credits from the student's balance.
6. System displays a submission confirmation with a reference ID.
7. At bid-window close, the allocation solver (UC-03: Run Allocation Solver) processes all submitted bids together, checking elective seat capacity and cross-referencing the Timetable System (UC-04: Check Timetable Collision, included) to eliminate schedule clashes.
8. System generates a proposed allocation list and forwards it to the Academic Registrar for review (UC-05).
9. Academic Registrar reviews and publishes the final allocation (UC-07).
10. System notifies the student of their final elective assignment via portal and email. Use case ends successfully.

Alternate Flow — 3a. Prerequisite Not Met

- 3a1. System detects that the student has not completed a prerequisite for one of the ranked electives.
- 3a2. System flags the ineligible elective in the ranking and displays a validation error naming the missing prerequisite.
- 3a3. Student either removes the ineligible elective and re-ranks, or replaces it with an eligible one, and resubmits.
- 3a4. If the student does not correct the bid before the window closes, only the validated (eligible) portion of the ranking is submitted to the solver.

Alternate Flow — 3b. Credit Allocation Invalid

- 3b1. System detects that the total distributed credits are not exactly 100, or that a duplicate elective rank was entered.
- 3b2. System rejects the submission and highlights the offending row(s) for correction.
- 3b3. Student adjusts the credit distribution and resubmits; flow resumes at Main Success Scenario step 3.